

100 Days of Azure DevOps

Day 60 Handout — Package once, run on one path you can pay for and tear down — slogans do not get invoices

6 - Containers & Kubernetes · Personal lab only · Educational content · Not a sales pitch

Architecture A — Phase 6 map (one artifact, one path)

Stage	Object	Done looks like
Pack	Dockerfile + layer cache + digest	Image builds locally and in CI
Store	ACR repository, tag = Build.BuildId	Not Hub; not latest-as-identity
Publish	Docker@2 on ubuntu-latest	Laptop is not the publisher
Run (pick one)	ACI or Container Apps or AKS+Helm/manifests	Pipeline applied it; Git knows the tag
Stop	Delete ACI group / CA app / AKS	Cost Analysis does not include a forgotten gym

Architecture B — pick one runtime, not a buffet

Constraint	Pick	Do not
Need a URL, no kube API	ACI or Container Apps	Stand up AKS 'so the recap looks senior'
Need kube + Helm	AKS with destroy date + helm upgrade --install	Three half-paths (ACI + CA + AKS) all 'almost'
Cost already bites	kind + ACR proof + skip Azure runtime	Leave node-count 1 because tomorrow-you might need it
Tag	Build.BuildId everywhere	Promote by latest in the recap diagram
Namespace	workloads	Anything in kube-system in the architecture one-liner

Architecture C — leftovers that still bill after a proud recap post

Leftover	What you see	Fix
ACI group	dns-name-label still resolves, meter runs	az container delete
aks-lab	System pool VM in Cost Analysis	az aks delete — the date from Day 56
Public IP / NSG orphans	RG not empty after cluster delete	az group delete rg-day56 if it was lab-only
ACR junk tags	Untagged manifests	Delete what the mini project does not need
Kubeconfig clutter	kubectl still points at a dead cluster	kubectl config delete-context

One-liner to remember

Phase 6 recap: package once, run anywhere — but 'anywhere' still has a bill. Done = image in ACR + one successful automated deploy path.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription and azure-100-labs. One path only. Recap one-liner goes in docs/phase6-recap.md.

- 1 Confirm the pipeline still builds and pushes myapp:\$(Build.BuildId) to ACR. Copy the digest into the recap doc.
- 2 Choose one run path: ACI, Container Apps, or AKS (manifests or Helm). Automate it from the pipeline or a documented CLI that uses the build tag. Not three paths.
- 3 Prove the running bits equal that digest/tag. If you skipped Azure runtime, prove kind ran the same tag and say so — skipping AKS is an architecture decision.

- 4 Tear down the run path: delete ACI/CA/AKS leftovers from Days 54–59. Empty RGs or document why ACR stays for Phase 7.
- 5 Write docs/phase6-recap.md: architecture one-liner, the path you picked, destroy receipts, kube-system vs workloads in one sentence.
- 6 LinkedIn recap from the personal account. No employer cluster diagrams. No buffet of half-finished runtimes.

Definition of done — one publisher, one tag, one path

```
# Mini project receipt (fill real names)
# ACR: <uniqueacr>.azurecr.io/myapp:<Build.BuildId> digest sha256:...
# Path: aci | container-apps | aks-helm | kind-only
# Apply: pipeline Docker@2 + (KubernetesManifest@1 | helm upgrade --install | az container create)
# Stop: az container delete | az aks delete | az containerapp delete
# Rule: workloads namespace, never kube-system; latest is not the promotion ID
```

Done checklist

- Image in ACR tagged with a build id, digest recorded
- Exactly one automated deploy path proved
- Paid runtimes from this phase deleted or dated
- docs/phase6-recap.md has the architecture one-liner

Tomorrow — Day 61

Entra ID fundamentals — the bouncer list every other control assumes